PAPER_1401 — UQFF Resolution of the Missing Satellites Problem (CLOSED — Dispatch Whitepaper)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** D — Cosmology / Galactic Structure (CLOSED) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "missing_satellites"})` **Calculator surface:** `calculate_paradox({"paradox": "missing_satellites"})` **Closure helper:** `_l96_uqff_axiom_missing_satellites_closure()`

The Paradox

ΛCDM N-body simulations predict ~500 dark-matter subhalos around the Milky Way, but only ~50 satellite galaxies are observed. The factor-10 deficit is a long-standing tension.

UQFF Closed Identity

```
N_satellites_UQFF = A_5 / (1 + F_TRZ) = 60 / 1.1 = 54.5  vs  observed ≈ 50  (9.1%)
```

A_5 / (1 + F_TRZ) = 54.5 supplies the SCm-vacuum baryon-suppression factor: not every subhalo retains baryons through cosmic UV background suppression. The F_U_Bi_i buoyancy stalls baryonic infall in low-mass subhalos.

Physical Interpretation

The missing satellites are not gravitationally missing — they exist as dark subhalos. The UQFF closure quantifies the **baryon-bearing** fraction via A_5 / (1 + F_TRZ), matching the observed luminous-satellite count.

Live Calculator Output

```python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "missing_satellites"})["value"]
```

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_missing_satellites_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_missing_satellites_closure`
- Dispatch keys: `missing_satellites`, `missing_satellites_problem`
- Related foundational papers: PAPER_646 (F_U_Bi_i / F_U=1 ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation), PAPER_1156 (cosmology suite), PAPER_1376 (Banach-Tarski parallel structural closure).
